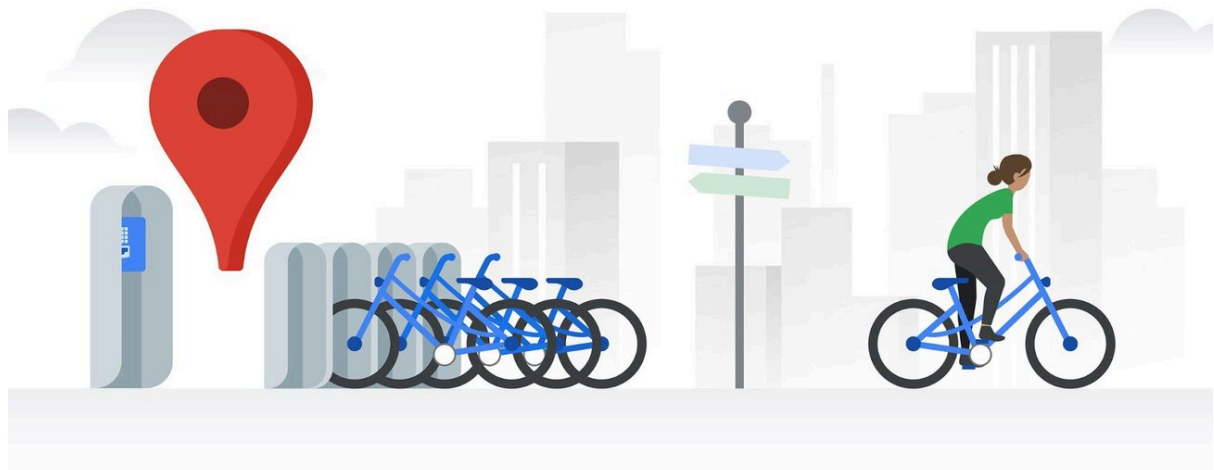


Google Data Analytics Certificate

Case study 1: How does a bike-share navigate speedy success?



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Date: May 2026

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1. Introduction (ask)

Cyclistic is a bike-share company based in Chicago. The company's business goal is to increase the number of annual memberships by converting casual riders into members.

The analysis aims to support this goal by identifying behavioral differences between casual riders and annual members, in order to inform data-driven marketing strategies.

The key question guiding this analysis is: how do annual members and casual riders use Cyclistic bikes differently?

Understanding these differences will help identify usage patterns that may explain why casual riders are not converting into annual memberships.

Key stakeholders include Lily Moreno, the marketing director, who is responsible for developing campaigns and initiatives to grow the bike-share program, and the Cyclistic executive team, who will decide whether to approve the proposed marketing strategy.

2. Prepare

The data used in this analysis was provided by the Google Data Analytics Certificate and is stored in a public online bucket. The datasets are available as CSV files.

Initially, all files were downloaded for review. After exploring their structure, it was observed that the datasets were not fully consistent in format. While some files contained monthly trip records, others were structured as quarterly or historical aggregations.

To ensure the analysis was both relevant and representative, only the most recent 12 months of available data were selected (from 2025-05 to 2026-04). This timeframe

provides a complete annual cycle, allowing seasonal usage patterns to be captured while reflecting current user behavior. Additionally, limiting the dataset to one year helps maintain a manageable data volume without compromising analytical reliability.

During the initial data validation process, an inconsistency was identified between one folder name and the CSV filename it contained. While the folder was labeled “202601”, the corresponding CSV file was named “202501”. As a result, it was unclear whether the data corresponded to the year 2026 or 2025. To verify the correct temporal scope of the dataset, the date range was checked in Google BigQuery using the trip start timestamps. The analysis confirmed that the data primarily corresponded to January 2026, although a small number of records dated 2025-12-31 were also present, likely because some trips began before midnight and ended after the start of the new year. A further verification showed that only 23 trips corresponded to December 2025, while 137,764 rides belonged to January 2026. The dataset was therefore retained once the date range had been confirmed. The file was renamed from “202501” to “202601” for consistency purposes.

Since the dataset contained a very large volume of records covering an entire year of bike trips, Google BigQuery was selected as the main tool for data storage, validation, transformation, and analysis.

Since some CSV files exceeded BigQuery's local upload limit (100 MB), the datasets were first uploaded to Google Cloud Storage and then imported into BigQuery from the storage bucket. To ensure that all selected CSV files were suitable for analysis, each file was uploaded to Google BigQuery as an independent table.

First, schema consistency was verified across all files. All datasets shared identical column names and data types, allowing them to be safely merged into a single table containing **5,697,445 trip records**. Once schema consistency was confirmed, the datasets were merged into a single consolidated table for further analysis.

3.Process

After importing and combining the 12 monthly datasets into a unified annual table, a series of data quality checks were conducted.

Duplicate checks were performed using the “ride_id” field, and no duplicate records were found.

Missing value analysis revealed null values only in station-related variables: **1,215,745** missing values in “start_station_name” and **1,279,863** in “end_station_name”. Since these fields were not essential for the overall behavioural analysis, these records were retained and excluded only from station-specific analyses.

Trip duration was then examined to identify anomalous records. An initial inspection based on minute-level calculations suggested a large number of zero-duration trips. However, further analysis at the second level showed that most of these records corresponded to legitimate very short rides lasting less than one minute, which had been rounded down by SQL time functions.

Only trips with clearly invalid durations were removed, including:

- **29 trips with negative duration**
- **1,777 trips with exactly zero seconds duration**

A small number of trips longer than 24 hours were also identified. Although initially flagged as potential anomalies, further inspection showed that most lasted between **25 and 26 hours** and displayed plausible characteristics (mainly “classic_bike” rides involving both casual users and members). Since no evidence suggested data corruption or recording errors, these records were retained in the dataset.

In total, **1,806 invalid records** were excluded, representing only **0.03%** of the original dataset. This cleaning process improved data reliability while preserving nearly all valid observations for subsequent analysis.

To facilitate the analysis, additional derived variables were created from the original timestamp fields.

First, trip duration was calculated in minutes (`ride_length`) using the difference between `ended_at` and `started_at`, allowing the comparison of travel time patterns between user types.

Next, the day of the week (`day_of_week`) was extracted from the trip start date to examine weekly usage patterns and identify differences between weekday and weekend behaviour.

Additionally, a month variable (`month_name`) and its corresponding numerical representation (`month_number`) were derived from the trip start timestamp. These variables were created to facilitate the analysis of seasonal ridership patterns and allow chronological visualisation of monthly trends.

Finally, the trip start hour (`start_hour`) was extracted to investigate hourly demand patterns and identify peak usage times.

These transformed variables made the dataset more suitable for exploratory analysis and enabled clearer comparisons between member and casual riders.

4. Analyze

4.1. Exploratory Data Analysis

After cleaning and preparing the dataset, exploratory analysis was conducted on 5,695,649 valid bike trips to identify behavioural differences between annual members and casual riders.

First, the distribution of user types was examined. The majority of trips (64.2%) were made by annual members (3,658,611 trips), while 35.8% corresponded to casual riders (2,037,038 trips), indicating that subscribed users represent the core customer base.

Significant differences were observed in average trip duration. Casual riders averaged 21.96 minutes per trip, almost twice as long as members, whose average trip duration was 11.94 minutes. This suggests that casual riders are more likely to use the service for leisure, tourism, or recreational activities, whereas members primarily rely on bicycles for shorter, practical journeys.

Weekly usage patterns further reinforced this distinction. Casual riders showed clear peaks on Saturday (416,420 trips) and Sunday (340,051 trips), indicating stronger weekend usage. In contrast, annual members recorded their highest activity during weekdays, particularly on Thursday (601,927 trips), with consistently high ridership from Monday to Friday. This pattern strongly suggests commuting behaviour among members.

Monthly ridership analysis revealed a strong seasonal pattern for both user groups. Trip volumes increased steadily from January onwards, reaching their highest levels during the summer months, particularly in July, August, and September. Although annual members consistently generated more trips throughout the year, casual riders exhibited a much stronger seasonal effect. Casual ridership increased from 24,733 trips in January to 337,740 trips in August, suggesting that leisure-oriented usage is highly influenced by favourable weather conditions and seasonal tourism. These findings indicate that seasonality affects all riders, but has a substantially greater impact on casual users.

Hourly trip distribution revealed additional differences between both user groups. Member trips displayed a clear bimodal commuting pattern, with strong peaks during morning hours (7–8 AM) and late afternoon (4–6 PM). The highest volume of member rides occurred at 5 PM, supporting the conclusion that members mainly use the service for daily transportation to and from work.

Casual riders, on the other hand, showed a more gradual increase in activity throughout the day, with usage concentrated between late morning and early evening, and no pronounced morning commuting peak. This suggests a more flexible and leisure-oriented riding behaviour.

Analysis of bicycle type preference showed a strong predominance of electric bikes, which accounted for 3,759,336 trips, compared to 1,936,313 trips made using classic

bikes. This indicates a clear user preference for electric bicycles, likely due to convenience and reduced physical effort.

Finally, the most frequently used starting stations included Navy Pier, DuSable Lake Shore Dr & Monroe St, Michigan Ave & Oak St, Millennium Park, and Theater on the Lake. Many of these locations are situated near major recreational and tourist attractions, reinforcing the hypothesis that casual riders frequently use the service for sightseeing and leisure purposes.

4.2. Key Insights

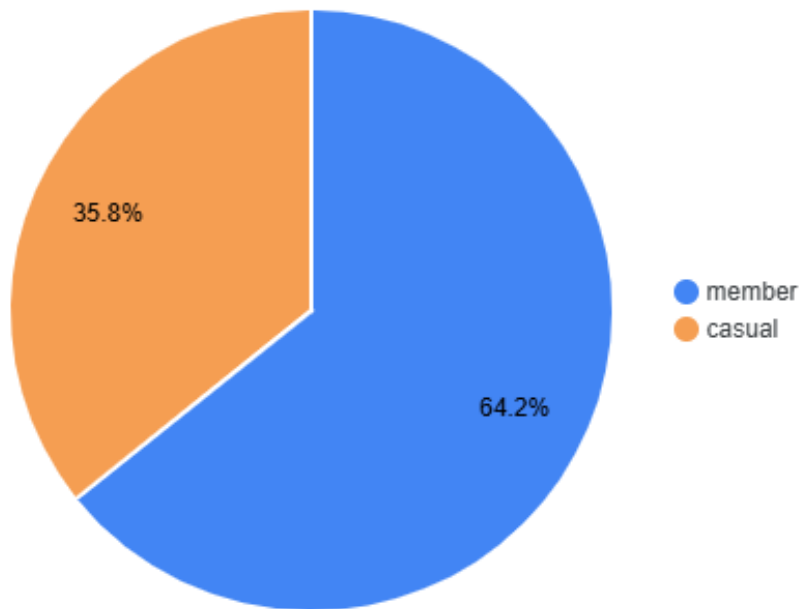
The analysis identified clear behavioural differences between annual members and casual riders:

- Members tend to use the bike-sharing service for short, routine, weekday trips, particularly during commuting hours.
- Casual riders tend to take longer rides, are more active on weekends, and frequently start trips near tourist and recreational areas.
- Electric bikes are the preferred option across the dataset.

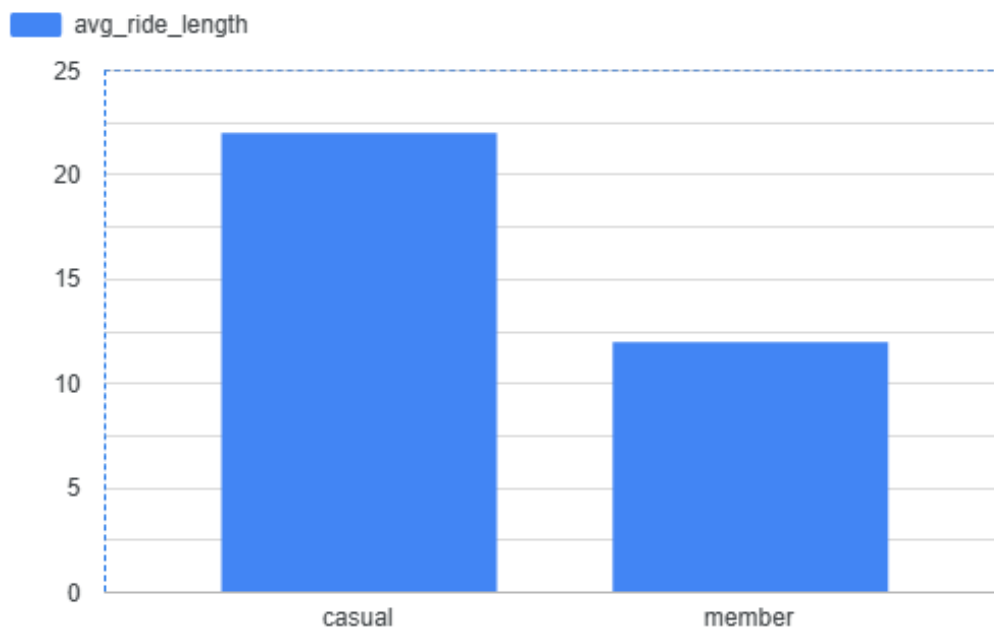
These findings suggest that casual riders may represent a valuable conversion opportunity, as their current usage patterns indicate engagement with the service but for different purposes than subscribed members.

5. Visualizations

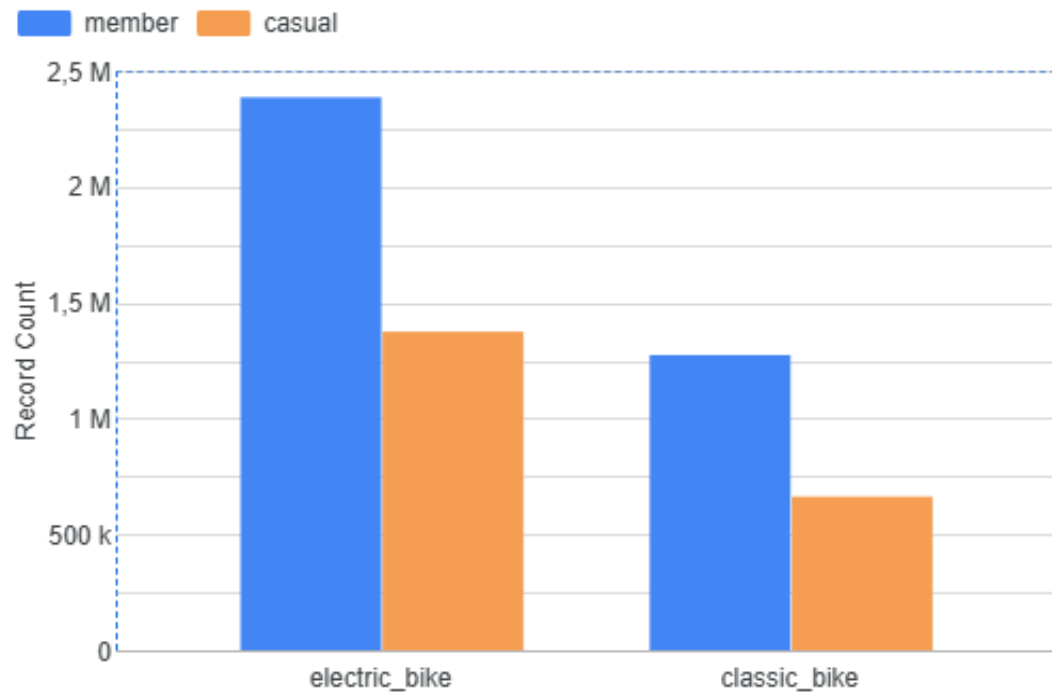
5.1. Total Member vs Casual



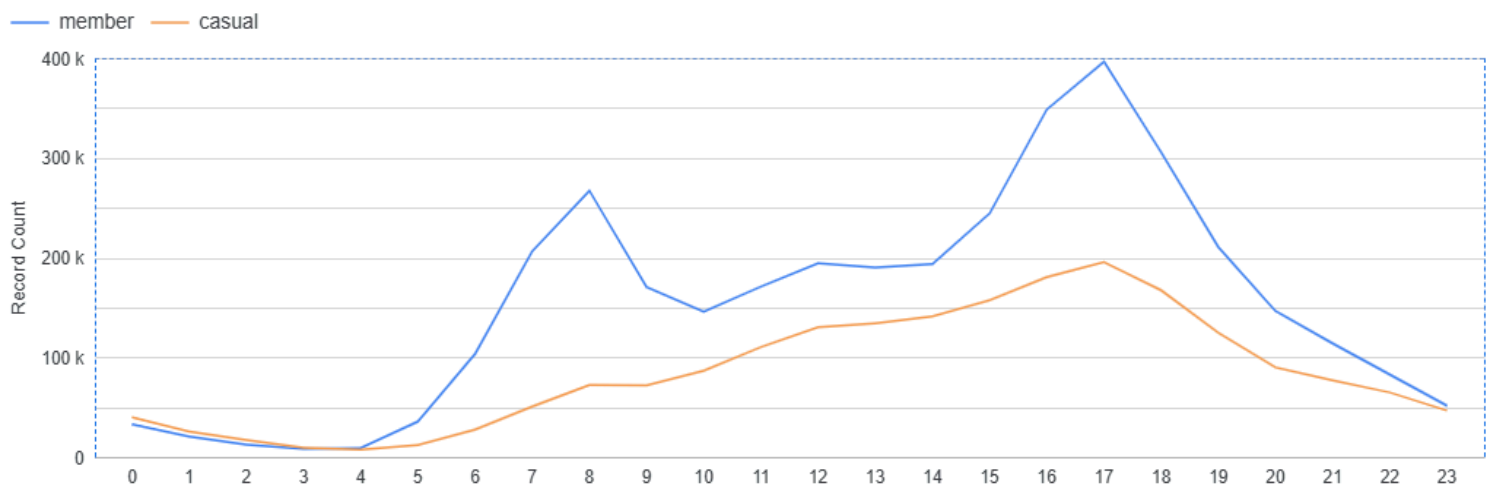
5.2. Average Ride Length (minutes)



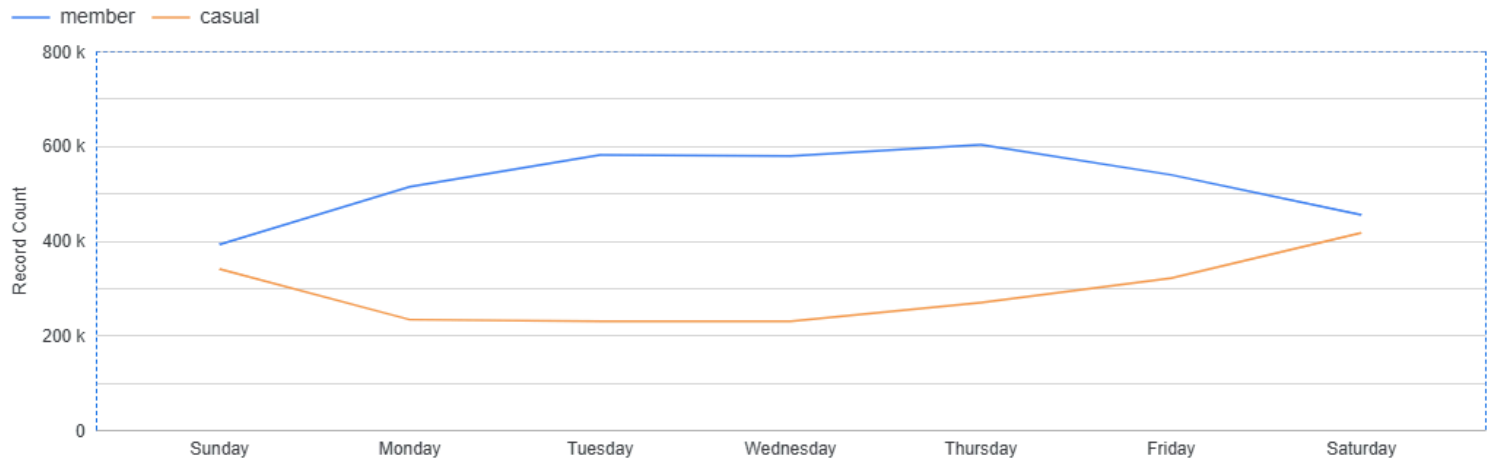
5.3. Bike Type Preference



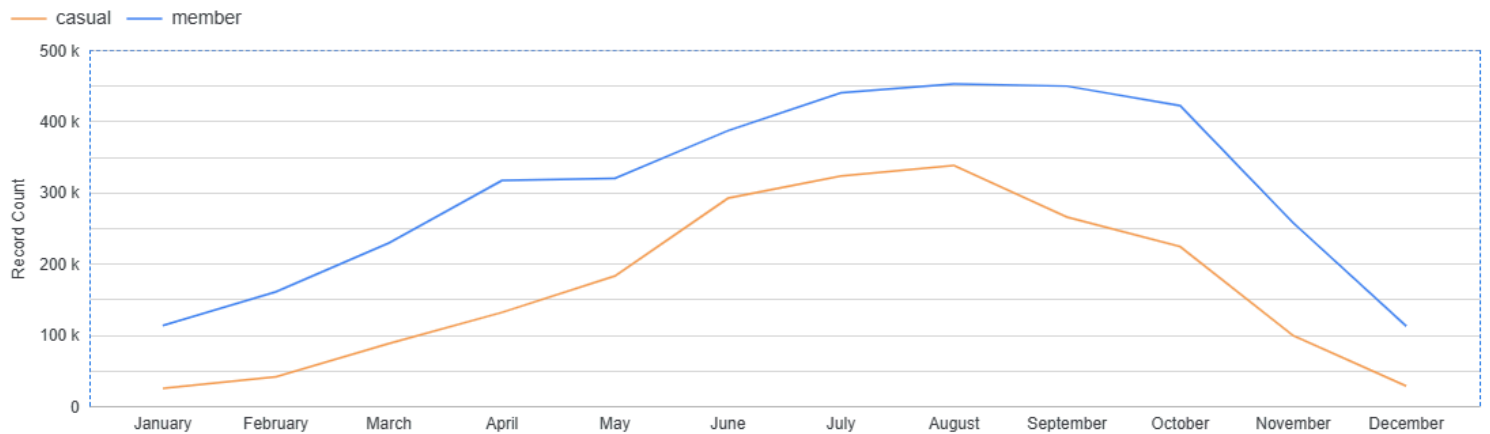
5.4. Trips by Hour



5.5. Trips by Day of Week



5.6. Trips by Month



6.Key Insights

The analysis revealed clear behavioural differences between annual members and casual riders.

First, annual members generated the majority of trips, accounting for approximately 64% of total rides. Their riding behaviour strongly suggests that bicycles are mainly used as a practical transportation method for commuting and daily mobility.

Members also showed significantly shorter average trip durations, with rides lasting approximately 12 minutes on average, compared to nearly 22 minutes for casual riders. This indicates that casual users are more likely to use the service for recreational, leisure, or tourism-related activities.

Temporal analysis further reinforced these differences. Annual members displayed clear commuting patterns, with strong usage peaks during weekday mornings and late afternoons, especially between 7–8 AM and 4–6 PM. In contrast, casual riders showed more flexible riding behaviour, with activity concentrated during afternoons, evenings, and weekends.

Weekly usage patterns highlighted that casual riders are substantially more active on Saturdays and Sundays, whereas members maintain consistently high usage throughout weekdays. This suggests that members primarily use bicycles for routine transportation, while casual riders use them more frequently for leisure purposes during free time.

Both user groups exhibited clear seasonal patterns, with ridership increasing steadily from spring and reaching its highest levels during the summer months. While annual members generated more trips throughout the year, both members and casual riders appeared to be influenced by seasonal factors such as weather conditions and outdoor activity levels.

The analysis also revealed a strong preference for electric bicycles, which accounted for the majority of trips across the dataset. This suggests that users value convenience and ease of use when choosing bike types.

Finally, the most frequently used stations were located near recreational and tourist areas such as Navy Pier, Millennium Park, and DuSable Lake Shore Drive. This further supports the conclusion that a significant proportion of casual riders use the service for sightseeing and recreational activities rather than commuting.

7. Recommendations

Based on the findings obtained during the analysis phase, the following recommendations are proposed to support the conversion of casual riders into annual members.

Since casual riders show significantly higher activity during weekends, Cyclistic could implement targeted weekend membership campaigns. Promotional discounts, free trial periods, or weekend-oriented membership packages could encourage recreational users to transition into annual members.

Casual riders take considerably longer trips than members, suggesting recreational usage patterns. Cyclistic could promote membership plans by highlighting potential savings and convenience benefits for users who frequently take extended rides.

Electric bikes represent the majority of trips within the dataset, indicating strong customer preference. Cyclistic could leverage this preference by offering membership incentives linked to electric bike access, such as reduced pricing or priority availability for members.

Some casual riders also display recurring weekday and afternoon usage patterns, suggesting potential commuting behaviour. Cyclistic could target these users with personalized membership offers focused on cost savings, convenience, and daily mobility benefits.

Cyclistic should focus membership conversion campaigns during the spring and summer months, when casual rider activity is at its highest. Targeting users during peak riding season may increase the likelihood of converting recreational riders into annual members.

8. Conclusion

This analysis identified clear behavioural differences between annual members and casual riders. Members primarily use the service for routine transportation and commuting, whereas casual riders show stronger leisure, tourism, and seasonal usage patterns. These findings suggest that targeted marketing initiatives focused on weekend riders, long recreational trips, electric bike incentives, and seasonal campaigns may increase the conversion of casual riders into annual members.

Appendix A - Data validation queries

```
SELECT
```

```
    table_name,  
  
    column_name,  
  
    data_type
```

```
FROM `bike-trip-capstone.bike_trips.INFORMATION_SCHEMA.COLUMNS`
```

```
ORDER BY column_name, table_name;
```

```
CREATE OR REPLACE TABLE `bike-trip-capstone.bike_trips.full_year` AS
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202505`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202506`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202507`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202508`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202509`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202510`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202511`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202512`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202601`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202602`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202603`
```

```
UNION ALL
```

```
SELECT * FROM `bike-trip-capstone.bike_trips.202604`;
```

```
SELECT COUNT(*) AS total_rows
```

```
FROM `bike-trip-capstone.bike_trips.full_year`;
```

Appendix B - Data cleaning queries

```
SELECT
```

```
    ride_id,
```

```
    COUNT(*) AS duplicates
```

```
FROM `bike-trip-capstone.bike_trips.full_year`
```

```
GROUP BY ride_id
```

```
HAVING COUNT(*) > 1;
```

```
SELECT
```

```
    COUNTIF(ride_id IS NULL) AS missing_ride_id,
```

```
    COUNTIF(started_at IS NULL) AS missing_started_at,
```

```
    COUNTIF(ended_at IS NULL) AS missing_ended_at,
```

```
    COUNTIF(start_station_name IS NULL) AS missing_start_station,
```

```
    COUNTIF(end_station_name IS NULL) AS missing_end_station
```

```
FROM `bike-trip-capstone.bike_trips.full_year`;
```



```
SELECT *  
  
FROM `bike-trip-capstone.bike_trips.full_year`  
  
WHERE ended_at < started_at;
```

```
SELECT  
  
    ride_id,  
  
    TIMESTAMP_DIFF(ended_at, started_at, MINUTE) AS ride_length  
  
FROM `bike-trip-capstone.bike_trips.full_year`  
  
ORDER BY ride_length;
```

```
SELECT  
  
    COUNT(*) AS total_0_duration_trips  
  
FROM  
  
    `bike-trip-capstone.bike_trips.full_year`  
  
WHERE  
  
    TIMESTAMP_DIFF(ended_at, started_at, MINUTE) = 0;
```

```
SELECT  
  
    TIMESTAMP_DIFF(ended_at, started_at, SECOND) AS trip_seconds,  
  
    COUNT(*) AS trips  
  
FROM `bike-trip-capstone.bike_trips.full_year`  
  
WHERE TIMESTAMP_DIFF(ended_at, started_at, MINUTE) = 0  
  
GROUP BY trip_seconds  
  
ORDER BY trip_seconds;
```

```
SELECT  
  
    ride_id,
```

```
rideable_type,  
member_casual,  
  
TIMESTAMP_DIFF(ended_at, started_at, HOUR) AS trip_hours  
FROM `bike-trip-capstone.bike_trips.full_year`  
  
WHERE TIMESTAMP_DIFF(ended_at, started_at, HOUR) > 24  
  
ORDER BY trip_hours DESC  
  
LIMIT 50;  
  
CREATE OR REPLACE TABLE `bike-trip-capstone.bike_trips.cleaned_data` AS  
SELECT *  
FROM `bike-trip-capstone.bike_trips.full_year`  
WHERE TIMESTAMP_DIFF(ended_at, started_at, SECOND) > 0;  
  
CREATE OR REPLACE TABLE `bike-trip-capstone.bike_trips.analysis_ready` AS  
SELECT  
*,  
  
TIMESTAMP_DIFF(ended_at, started_at, MINUTE) AS ride_length_minutes,  
  
FORMAT_DATE('%A', DATE(started_at)) AS day_of_week,  
  
FORMAT_DATE('%B', DATE(started_at)) AS month_name,  
  
EXTRACT(MONTH FROM started_at) AS month_number,  
  
EXTRACT(HOUR FROM started_at) AS start_hour,  
  
EXTRACT(DAYOFWEEK FROM started_at) AS day_number  
FROM `bike-trip-capstone.bike_trips.cleaned_data`;
```

Appendix C – Exploratory analysis queries

```
SELECT COUNT(*)  
FROM `bike-trip-capstone.bike_trips.analysis_ready`;
```

```
SELECT
    member_casual,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY member_casual;
```

```
SELECT
    member_casual,
    AVG(ride_length_minutes) AS avg_trip_duration
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY member_casual;
```

```
SELECT
    day_of_week,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY day_of_week
ORDER BY trips DESC;
```

```
SELECT
    start_hour,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY start_hour
ORDER BY start_hour;
```

```
SELECT
    member_casual,
    month_name,
    month_number,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY
    member_casual,
    month_name,
    month_number
ORDER BY
    member_casual,
    month_number;
```

```
SELECT
    rideable_type,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY rideable_type;
```

```
SELECT
    start_station_name,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY start_station_name
ORDER BY trips DESC
LIMIT 10;
```

```
SELECT
    member_casual,
    day_of_week,
    day_number,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY member_casual, day_of_week, day_number
ORDER BY member_casual, day_number;
```

```
SELECT
    start_station_name,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
WHERE start_station_name IS NOT NULL
GROUP BY start_station_name
ORDER BY trips DESC
LIMIT 10;
```

```
SELECT
    member_casual,
    start_hour,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY member_casual, start_hour
ORDER BY member_casual, start_hour;
```

```
SELECT
    member_casual,
    rideable_type,
    COUNT(*) AS trips
FROM `bike-trip-capstone.bike_trips.analysis_ready`
GROUP BY member_casual, rideable_type;
```